

Alessio Salvatore

 Github |  alessiosalvatore1703@gmail.com |  LinkedIn |  Website
 Italian |  March 17, 2003 |  +39 3339858633

Work Experience

- Laelaps AI** Jun 2026–Present
Robotics Engineer
- Built from scratch a multi-agent RL pipeline teaching robot teams to patrol large areas autonomously, from the simulator to the cloud training pipeline
 - Defined metrics to evaluate the quality of the patrol, its adaptability to changes in the environment, and its unpredictability
 - Deployed and evaluated the trained policies on Unitree Go2 quadrupeds, scoring my framework against state-of-the-art approaches; submitted as a paper to ICRA 2027
- ETH Robotics Club** Nov 2025–Aug 2026
Partnerships Team Lead. Closed the club's 2025–2026 partnerships, including the annual sponsor and Hack2026
- JOINT Lab — Italian Institute of Technology, Bergamo**  Repo Feb–Jun 2025
Research Intern — Industrial Robotics. Linear actuator CAD, real-time Arduino control, MATLAB validation

Education

- ETH Zürich, Zürich, Switzerland** Sep 2025–Present
M.Sc. in Robotics
Coursework: Probabilistic AI, Advanced Machine Learning, Deep Generative Models, Robot Learning, Perception and Learning for Robotics, 3D Vision, Image Analysis and Computer Vision, Planning and Decision Making
- Politecnico di Milano, Milan, Italy** Sep 2022–Jul 2025
B.Sc. in Automation Engineering
Final Grade: 110/110 *cum laude* (equivalent GPA $\approx$ 4.0/4.0; 29.57/30).
- Illinois Institute of Technology, Chicago, IL, USA** Aug–Dec 2024
Exchange — Global E3 Program

Projects

Research Projects

- Knowledge Distillation for Aerial Robots Control**  Video  Report Feb–Jun 2026
Autonomous Systems Lab
- Model-based controllers for an overactuated drone (OMAV) lose accuracy when the vehicle drifts from its model, and disturbance observers can be too slow to compensate
 - Distilled multiple expert controllers, each tuned to a different version of the vehicle, into a single recurrent neural controller that adapts from its recent flight history: in simulation it tracks velocity better and diverges far less under $\pm 30\%$ model errors (41.5% $\rightarrow$ 24%), and it flew the OMAV without retraining
- Goal-Driven Adaptive-Resolution 3D Mapping with RL**  Report Feb–Jun 2026
Computer Vision and Geometry Group
- A robot mapping a building at uniform voxel resolution spends most of its memory on regions irrelevant to its task
 - Trained a PPO agent reallocating resolution per region online under a memory budget, on a sparse hierarchical grid in Habitat/HM3D: it keeps 79% of a uniformly fine map's point-goal success with 87% less memory on 20 unseen episodes

Selected Projects

- Autonomous Exploration on a Unitree A2 Quadruped**  Repo 06/2026
ETH Robotics Summer School (ETHZ-RobotX) — deployed an autonomous search-and-explore mission with our team on a Unitree A2 at three field sites
- Vision-Language-Action Policies on an SO-101 Arm**  Repo 05/2026
Project of the Class Robot Learning — fine-tuned SmoVLA for language-conditioned pick-and-place on an SO-101 arm at 10 FPS
- Person Following on the Agibot X2 Humanoid**  Repo 05/2026
RoboHack Hackathon — built real-time onboard person detection and following in a hackathon weekend: YOLOv8 stereo-disparity depth, Ruckig head tracking, and a state-machine locomotion supervisor holding a set distance